



# Modelling Financial Markets

## Seminar 2: Market Efficiency — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, [antoaneta.amza@csie.ase.ro](mailto:antoaneta.amza@csie.ase.ro)

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

## Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **AMH:** Adaptive Markets Hypothesis
- ▣ **AR:** AutoRegressive (model); in event studies: Abnormal Return
- ▣ **BDS:** Brock–Dechert–Scheinkman (test)
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BET-TR:** BET Total Return (index)
- ▣ **BIST:** Borsa İstanbul (BIST 100 index) — *the Istanbul Stock Exchange (BIST 100 index)*
- ▣ **BUX:** Budapesti Értéktőzsde index — *the Budapest Stock Exchange index*
- ▣ **BVB:** Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CD:** Chow–Denning (multiple variance-ratio test)
- ▣ **CI:** Confidence Interval
- ▣ **COVID-19:** COronaVirus Disease 2019
- ▣ **CV:** Cross-Validation
- ▣ **DFA:** Detrended Fluctuation Analysis
- ▣ **ETF:** Exchange-Traded Fund
- ▣ **FRED:** Federal Reserve Economic Data
- ▣ **FTSE:** Financial Times Stock Exchange (FTSE Russell, index provider)

## Acronyms used in this material (2/2)

- ▣ **FWER**: Family-Wise Error Rate
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **IPC**: Índice de Precios y Cotizaciones — *the Mexican Stock Exchange index*
- ▣ **NW**: Newey–West (standard errors)
- ▣ **OLS**: Ordinary Least Squares
- ▣ **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ **RW1**: Random Walk 1 (i.i.d. increments)
- ▣ **RW2**: Random Walk 2 (independent increments)
- ▣ **RW3**: Random Walk 3 (uncorrelated increments)
- ▣ **SD**: Standard Deviation
- ▣ **SE**: Standard Error
- ▣ **TOM**: Turn Of the Month
- ▣ **US**: United States
- ▣ **VR**: Variance Ratio
- ▣ **WIG20**: Warszawski Indeks Giełdowy 20 — *the Warsaw Stock Exchange index of the 20 largest companies*

## Today's Question and Route

- Question of the seminar: **which weak-form rejections of Chapter 2 survive robust inference, several horizons and several tests?**
- Route
  - ▶ what you need for today: five primer slides (the lecture comes after this seminar)
  - ▶ A1, A5–A6: a variance ratio and an implied Hurst exponent on paper
  - ▶ B1, B7, C2: classical vs robust variance ratios, the crisis check, an AI answer to audit
  - ▶ B5, B6: a calendar effect with HAC errors, then fifteen tests with Holm
  - ▶ after the lecture: B3, B4, DFA and variance ratios on rolling windows with bootstrap bands
- What you produce: the A worksheets, one table and one figure per B exercise, the C2 audit, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section
- Seminar: Antoaneta Amza, PhD candidate, [antoaneta.amza@csie.ase.ro](mailto:antoaneta.amza@csie.ase.ro)

## Exercise Map

| Exercise          | Question                                                                                 | Type                                      | Model        |
|-------------------|------------------------------------------------------------------------------------------|-------------------------------------------|--------------|
| A1                | what does $\widehat{VR}(2)$ compute, line by line?                                       | Solved                                    | –            |
| A5, A6            | does an implied $H(20) \neq 0.5$ prove long memory?                                      | Solved,<br>Proposed                       | A5           |
| B1                | do the S&P 500 and BET reject RW3 with robust errors?                                    | Solved                                    | –            |
| B7                | is the S&P 500 reversal a crisis effect?                                                 | Proposed                                  | B1           |
| C2                | what is wrong in an AI variance-ratio answer?                                            | Proposed                                  | A1, B1       |
| B5                | is the turn-of-the-month effect real?                                                    | Solved                                    | –            |
| B6                | which of 15 calendar effects survive Holm?                                               | Proposed                                  | B5           |
| After the lecture | B3 [Solved], B4 [Proposed]: rolling DFA and rolling variance ratios with bootstrap bands | homework                                  | B1           |
| Extensions        | A2, A3, A4, B2, B8, B9, C1, C3 (optional, after the seminar)                             | Solved or<br>Proposed<br>(marked on each) | on each task |

- ▣ **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: derivations on paper; Part B: computation and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

## Data Used

| Series                        | Price used                                                                         | Calendar             | Used in                |
|-------------------------------|------------------------------------------------------------------------------------|----------------------|------------------------|
| S&P 500                       | close                                                                              | US trading days      | A6, B1, B3, B5–B7, C2  |
| BET                           | close                                                                              | BVB trading days     | A5, B1, B2, B5, B6, C1 |
| BET-TR and 8 emerging indices | close (WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai)                 | own calendar each    | B2, B6, C1             |
| Euro Stoxx 50                 | close                                                                              | own calendar         | B6                     |
| Bitcoin                       | close (USD)                                                                        | every day            | B4                     |
| Banca Transilvania, BRD       | adjusted close (dividends, splits)                                                 | common days with BET | B9                     |
| S&P 500 monthly               | Shiller S&P Composite, S&P 500 closes, SPY dividends; 3-month T-bill (FRED, TB3MS) | month ends           | B8                     |
| 49 industry portfolios        | <a href="#">Kenneth French Data Library</a> , value-weighted, monthly              | month ends           | C3                     |

- Daily closing prices of the course data, one series per asset, from 2000 (Bitcoin from 2011) to 18 Sep 2026; daily log returns  $r_t = \ln P_t - \ln P_{t-1}$  on each series' own calendar
- Cleaning rule for indices: weekdays only; a close equal to the previous close is removed (usually a filled holiday)

## Setup: Run and Verify

```
# notebook sections "Data" and
# "Efficiency tests" define
# log_returns(name, start=None)
# variance_ratio(r, q) -> (VR, z, z*)
# chow_denning(r), runs_test(r), dfa_hurst(r)
sp = log_returns('sp500') # S&P 500
bet = log_returns('bet') # BET
print(len(sp), sp.index[0].date(),
      sp.index[-1].date(), len(bet))
print(sp.head(3).round(5))
```

- Expected: 6 714 S&P 500 returns from 2000-01-04 to 2026-09-18; 6 670 BET returns

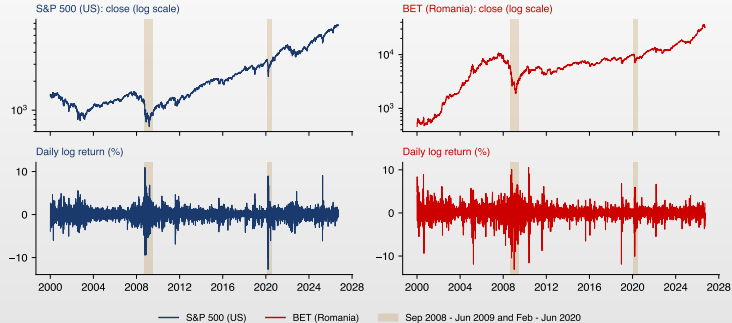
| date       | close   | $r_t$    |
|------------|---------|----------|
| 2000-01-03 | 1455.22 | —        |
| 2000-01-04 | 1399.42 | −0.03910 |
| 2000-01-05 | 1402.11 | +0.00192 |
| 2000-01-06 | 1403.45 | +0.00096 |

- If your count or first value differs, check the series, the price type (adjusted or unadjusted close) and the cleaning rule

## Notation Used in Every Exercise

- ▣  $r_t$ : daily log return;  $e_t = r_t - \bar{r}$ ;  $T$ : number of returns;  $q$ : horizon in days
- ▣  $\hat{\rho}_k = \sum_{t>k} e_t e_{t-k} / \sum_t e_t^2$ ; robust SE  $\sqrt{\hat{\tau}_k}$ ,  $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$
- ▣  $VR(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho_k$ ;  $z(q)$  assumes RW1 (i.i.d.),  $z^*(q)$  allows volatility clustering (RW3) [Lo & MacKinlay \(1988\)](#)
- ▣ Chow–Denning  $CD = \max_q |z^*(q)|$ ,  $q = 2, 5, 10, 20$ , 5% critical value 2.49 [Chow & Denning \(1993\)](#)
- ▣ Implied Hurst exponent  $H(q) = 0.5 + \ln VR(q) / (2 \ln q)$ , exact only for self-similar increments; DFA exponent  $\alpha_{DFA}$  [Peng et al. \(1994\)](#)
- ▣ HAC: Newey–West errors [Newey & West \(1987\)](#); wild bootstrap: multiply  $e_t$  by random signs  $\pm 1$  [Kim \(2006\)](#); FWER: probability of at least one false rejection

## The Data at a Glance



- Volatility clusters: calm years and bursts (2008–2009, 2020) alternate, so i.i.d. standard errors are too small in every B exercise

## What You Need for Today (1/5): Weak-Form Efficiency and the Random Walk

- This seminar comes **before** the lecture: these five slides give only what the in-class tasks use; depth in [Chapter 2](#), the lecture after this seminar
- **Weak-form efficiency** [Fama \(1970\)](#): past prices and returns cannot be used to earn risk-adjusted profits; testable consequence: returns are not predictable from their own past
- **Random walk** of the log price,  $\ln P_t = \mu + \ln P_{t-1} + \varepsilon_t$ , in three versions [Campbell, Lo & MacKinlay \(1997\)](#)
  - ▶ RW1:  $\varepsilon_t$  i.i.d.; RW2: independent, the variance may change; RW3: only uncorrelated,  $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$ , while  $\varepsilon_t^2$  may be autocorrelated
  - ▶  $\mu$ : the drift, the expected daily log return;  $\varepsilon_t$ : the shock of day  $t$ , with mean 0
  - ▶ [Chapter 1](#) showed volatility clustering, so RW1 fails before any test: the relevant null for daily returns is RW3
- Sign of the dependence:  $\rho_k > 0$  is persistence (momentum),  $\rho_k < 0$  is reversal (mean reversion); a significant  $\hat{\rho}_k$  is not yet a profitable rule (costs, risk)
- From [Chapter 1](#) you already have  $\hat{\rho}_k$ , the i.i.d. band  $\pm 1.96/\sqrt{T}$  and the robust standard error  $\sqrt{\hat{\tau}_k}$  (Notation slide)

## What You Need for Today (2/5): The Variance Ratio

- Uncorrelated returns with constant variance: the variance of a  $q$ -day sum is  $q$  times the one-day variance
- **Variance ratio Lo & MacKinlay (1988):**  $VR(q) = \frac{\text{Var}(r_t + \dots + r_{t-q+1})}{q \text{Var}(r_t)}$ ; it equals 1 under RW1, RW2 and RW3
  - ▶ expanding the variance of the sum:  $VR(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho_k$ , a weighted sum of autocorrelations; in particular  $VR(2) = 1 + \rho_1$
  - ▶  $VR(q) > 1$ : persistence;  $VR(q) < 1$ : reversal
- Example:  $\rho_1 = -0.1$ , all other  $\rho_k = 0$ :  $VR(2) = 0.9$ ;  $VR(5) = 1 + 2(1 - 1/5)(-0.1) = 0.84$
- Lo-MacKinlay estimator (B1): overlapping  $q$ -day sums with bias corrections
  - ▶  $\hat{\sigma}_a^2 = \frac{1}{T-1} \sum_t e_t^2$ ,  $\hat{\sigma}_c^2(q) = \frac{1}{m} \sum_{t=q}^T \left( \sum_{j=0}^{q-1} r_{t-j} - q\hat{\mu} \right)^2$ ,  $m = q(T - q + 1)(1 - q/T)$ ,  $\widehat{VR}(q) = \hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$
  - ▶  $\hat{\mu}$ : mean daily return;  $\hat{\sigma}_a^2$ : one-day variance;  $\hat{\sigma}_c^2(q)$ : variance of the overlapping  $q$ -day sums divided by  $q$ ;  $m$ : a divisor that removes the small-sample bias

## What You Need for Today (3/5): Testing $VR(q) = 1$

- Under RW1:  $z(q) = (\widehat{VR}(q) - 1) / \sqrt{2(2q - 1)(q - 1)/(3qT)} \approx N(0, 1)$ ; for  $q = 2$  the denominator is  $1/\sqrt{T}$
- Under RW3 (volatility clustering allowed): each  $\hat{\rho}_k$  has variance  $\hat{\delta}_k/T$ , the robust variance of [Chapter 1](#) ( $\hat{\delta}_k = T\hat{\tau}_k$ )
  - ▶  $\hat{\delta}_k = T \sum_{t>k} e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$ ,  $\hat{\theta}(q) = \sum_{k=1}^{q-1} (2(q-k)/q)^2 \hat{\delta}_k$ ,  $z^*(q) = \sqrt{T} (\widehat{VR}(q) - 1) / \sqrt{\hat{\theta}(q)}$
  - ▶  $\hat{\theta}(q)$ : the robust variance of  $\sqrt{T} \widehat{VR}(q)$ , the weights  $2(q-k)/q$  being those of  $\rho_k$  in  $VR(q)$ ;  $|z^*| > 1.96$  rejects at 5%
  - ▶ under i.i.d. returns  $\hat{\delta}_k \approx 1$  and  $z^* \approx z$ ; under clustering  $\hat{\delta}_k > 1$  and  $|z^*| < |z|$
- Example:  $T = 2500$ ,  $\widehat{VR}(2) = 0.95$ :  $z(2) = -0.05 \times 50 = -2.5$ , a rejection at 5%; with  $\hat{\delta}_1 = 4$ ,  $z^*(2) = -2.5/2 = -1.25$ , no rejection
- **Chow–Denning** [Chow & Denning \(1993\)](#): testing four horizons at once,  $CD = \max_q |z^*(q)|$ ,  $q = 2, 5, 10, 20$ , is compared with 2.49, not with 1.96, so that the joint test keeps a 5% level

## What You Need for Today (4/5): The Hurst Exponent

- **Hurst exponent**  $H \in (0, 1)$ , from the hydrology of the Nile [Hurst \(1951\)](#): how fast the spread of a cumulated series grows with the horizon
- Self-similar increments (fractional Gaussian noise):  $\text{Var}(r_t + \dots + r_{t-q+1}) = q^{2H} \text{Var}(r_t)$ , so  $\text{VR}(q) = q^{2H-1}$ 
  - ▶  $H = 0.5$ : no memory;  $H > 0.5$ : persistence that decays slowly (long memory);  $H < 0.5$ : anti-persistence
- Implied exponent at one horizon:  $H(q) = 0.5 + \ln \text{VR}(q) / (2 \ln q)$ ; example  $\text{VR}(20) = 1.2$ :  
 $H(20) = 0.5 + 0.182/5.991 = 0.530$
- Short-memory benchmark, AR(1) (first-order autoregression):  $r_t = \phi r_{t-1} + u_t$ ,  $\rho_k = \phi^k$ 
  - ▶  $\phi$ : the lag-1 autocorrelation,  $|\phi| < 1$ ;  $u_t$ : an i.i.d. shock; the dependence dies out geometrically (short memory)
  - ▶ its  $\text{VR}(q)$  follows from the weighted-sum formula and stays bounded as  $q$  grows, so  $H(q)$  drifts back to 0.5 (A5, A6)

## What You Need for Today (5/5): Calendar Effects and Multiple Testing

- **Turn of the month (TOM)** [Ariel \(1987\)](#): the last trading day of a month and the first three of the next; test  $r_t = a + b \text{TOM}_t + u_t$  by OLS (ordinary least squares)
  - ▶  $\text{TOM}_t = 1$  on a turn-of-the-month day, 0 otherwise;  $u_t$ : the residual; with this 0/1 dummy,  $\hat{a}$  is the mean on other days and  $\hat{b}$  the difference of the two means
  - ▶ **HAC** (heteroskedasticity- and autocorrelation-consistent) errors of [Newey & West \(1987\)](#): the variance of  $\hat{b}$  allowing changing variance and correlation up to  $L$  lags, with weights  $1 - j/(L + 1)$ ; in Python `cov_type='HAC'`
- **FWER** (family-wise error rate): the probability of at least one false rejection among  $M$  tests; independent tests at 5%:  $1 - 0.95^M$ ,  $M = 10$ : 40.1%;  $p_i$ : the  $p$ -value of test  $i$ ,  $\alpha$ : the target FWER
- **Bonferroni**: reject if  $p_i < \alpha/M$ ; **Holm Holm (1979)**: sort  $p_{(1)} \leq \dots \leq p_{(M)}$ , reject while  $p_{(j)} < \alpha/(M - j + 1)$ , stop at the first failure; both keep  $\text{FWER} \leq \alpha$
- Example,  $M = 4$ :  $p = 0.001, 0.015, 0.020, 0.300$ ; Holm thresholds 0.0125, 0.0167, 0.025, 0.05: three rejections; Bonferroni (0.0125): one
- After the lecture: DFA and bootstrap bands on rolling windows (B3, B4)

## A1 [Solved]: The Variance Ratio, Line by Line — Task

### ▣ Questions

- ▶ what does  $\widehat{VR}(2)$  compute?
- ▶ how does it relate to  $\hat{\rho}_1$ ?

- ▣ **Data:** eight daily returns in %, in time order: 1.0; -0.5; 0.8; 0.3; -1.2; 0.6; 0.9; -0.4; no other data; simple estimator, sums of squares divided by the number of terms

### ▣ Tasks

1. Compute the mean  $\hat{\mu}$ , the deviations  $e_t = r_t - \hat{\mu}$  and the one-day variance  $\hat{\sigma}_1^2 = \frac{1}{8} \sum_t e_t^2$ .
2. Compute the lag-1 autocorrelation  $\hat{\rho}_1 = \sum_{t=2}^8 e_t e_{t-1} / \sum_{t=1}^8 e_t^2$ .
3. Form the seven overlapping two-day sums  $s_t = r_{t-1} + r_t$  and compute  $\hat{\sigma}_2^2 = \frac{1}{7} \sum_t (s_t - 2\hat{\mu})^2$ .
4. Compute  $\widehat{VR}(2) = \hat{\sigma}_2^2 / (2\hat{\sigma}_1^2)$  and compare it with  $1 + \hat{\rho}_1$ .

- ▣ **Report:** a worksheet with columns  $t$ ,  $r_t$ ,  $e_t$ ,  $e_t^2$ ,  $e_t e_{t-1}$ ,  $s_t$ ,  $(s_t - 2\hat{\mu})^2$  and one sentence on the sign of  $\widehat{VR}(2) - 1$

- ▣ Notebook: section A1

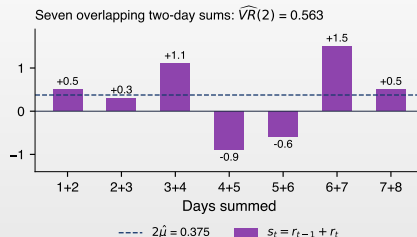
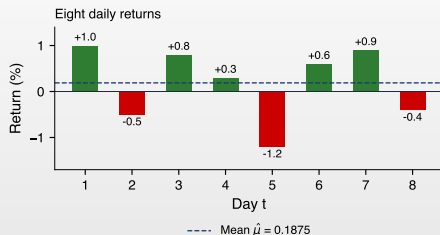
## A1 [Solved]: Solution (1/2) — Worksheet

| $t$ | $r_t$ | $e_t$   | $e_t^2$ | $e_t e_{t-1}$ | $s_t$ | $(s_t - 2\hat{\mu})^2$ |
|-----|-------|---------|---------|---------------|-------|------------------------|
| 1   | +1.0  | +0.8125 | 0.6602  | —             | —     | —                      |
| 2   | −0.5  | −0.6875 | 0.4727  | −0.5586       | +0.5  | 0.0156                 |
| 3   | +0.8  | +0.6125 | 0.3752  | −0.4211       | +0.3  | 0.0056                 |
| 4   | +0.3  | +0.1125 | 0.0127  | +0.0689       | +1.1  | 0.5256                 |
| 5   | −1.2  | −1.3875 | 1.9252  | −0.1561       | −0.9  | 1.6256                 |
| 6   | +0.6  | +0.4125 | 0.1702  | −0.5723       | −0.6  | 0.9506                 |
| 7   | +0.9  | +0.7125 | 0.5077  | +0.2939       | +1.5  | 1.2656                 |
| 8   | −0.4  | −0.5875 | 0.3452  | −0.4186       | +0.5  | 0.0156                 |
| sum |       |         | 4.4688  | −1.7639       |       | 4.4044                 |

$$\hat{\mu} = 1.5/8 = 0.1875; \hat{\sigma}_1^2 = 4.4688/8 = 0.5586; \hat{\rho}_1 = -1.7639/4.4688 = -0.395$$

$$\hat{\sigma}_2^2 = 4.4044/7 = 0.6292; \widehat{VR}(2) = 0.6292/(2 \times 0.5586) = 0.563, \text{ against } 1 + \hat{\rho}_1 = 0.605$$

## A1 [Solved]: Solution (2/2) — Chart and Interpretation

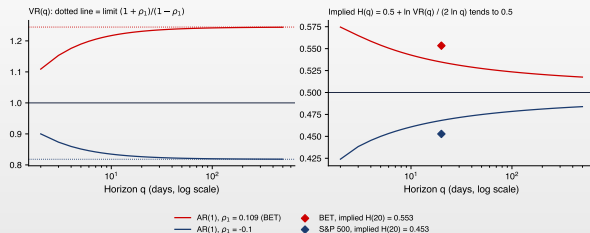


- $VR(2) = 1 + \rho_1$  holds for population moments; in a sample the two differ by end terms and divisors (7 sums, 8 squares): 0.563 vs 0.605 is a convention gap, not an error
- Interpretation: consecutive returns reverse, so two-day sums vary less than two single days: mean reversion. With  $T = 8$  this is arithmetic, not evidence: no test is attempted

## A5 [Solved]: The Implied Hurst Exponent of BET — Task

- ▣ **Question:** does BET's  $\widehat{VR}(20) > 1$  imply long memory?
- ▣ **Data:** BET, daily log returns 2000–2026 (closing prices):  $\widehat{VR}(20) = 1.377$ ,  $\hat{\rho}_1 = 0.109$  (Lecture 2); for exactly self-similar increments (fractional Gaussian noise)  $VR(q) = q^{2H-1}$
- ▣ **Tasks**
  1. Take logarithms of  $VR(q) = q^{2H-1}$  and solve for the implied  $H(q)$ .
  2. Compute the implied  $H(20)$  for BET.
  3. Compute  $VR(20)$  and  $H(20)$  of a short-memory AR(1) with  $\rho_k = 0.109^k$ , using 
$$VR(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho_k.$$
- ▣ **Report:** the formula, the two  $H(20)$  values with three decimals, and whether one horizon can identify long memory
- ▣ Notebook: section A5

## A5 [Solved]: Solution



- (1)  $\ln VR(q) = (2H - 1) \ln q$ , so  
 $H(q) = 0.5 + \ln VR(q) / (2 \ln q)$
- (2)  $\ln 1.377 = 0.3199$ ,  $2 \ln 20 = 5.991$ :  
 $H(20) = 0.5 + 0.3199/5.991 = 0.553$
- (3) AR(1):  $VR(20) = 1.231$ ,  $H(20) = 0.535$ , and  
 $H(250) = 0.520$  (red curve)
- Interpretation: short memory alone gives most of  
 BET's  $H(20)$ , and its  $H(q)$  drifts back to 0.5: one  
 horizon cannot identify long memory

## A6 [Proposed]: The Implied Hurst Exponent of the S&P 500 — Task

- ▣ **Question:** is the S&P 500 “anti-persistent”, or only short-run reversing?
- ▣ **Data:** S&P 500, daily log returns 2000–2026 (closing prices):  $\widehat{VR}(20) = 0.753$ ; a short-memory benchmark AR(1) with  $\rho_k = (-0.1)^k$
- ▣ **Tasks**
  1. Compute the implied  $H(20)$  of the S&P 500 with the formula of A5.
  2. Compute  $VR(20)$  and  $H(20)$  of the AR(1) benchmark.
  3. Compute  $VR(250)$ ,  $H(250)$  and the limit  $VR(\infty) = (1 + \rho)/(1 - \rho)$  of the AR(1), and read the blue curve of the A5 chart.
- ▣ **Report:** the five numbers and one sentence: does  $H(20) < 0.5$  require long memory?
- ▣ **Model:** A5 [Solved]
- ▣ Notebook: section A6

## B1 [Solved]: Classical vs Robust Variance Ratio — Task

- **Question:** do the S&P 500 and BET reject the random walk once volatility clustering is allowed?
- **Data:** S&P 500 and BET, daily log returns 2000–2026,  $T = 6\,714$  and  $6\,670$ ;  $H_0$  of  $z(q)$ : RW1 (i.i.d.);  $H_0$  of  $z^*(q)$ : RW3 plus the conditions of [Lo & MacKinlay \(1988\)](#)
- **Tasks**
  1. For the S&P 500 and  $q = 2$ , compute  $\hat{\sigma}_a^2$ ,  $\hat{\sigma}_c^2(2)$ ,  $\widehat{VR}(2)$ ,  $\hat{\delta}_1$ ,  $z(2)$  and  $z^*(2)$  with the Lo–MacKinlay formulas (next slide).
  2. For both markets and  $q = 2, 5, 10, 20$ , compute  $\widehat{VR}(q)$ ,  $z(q)$ ,  $z^*(q)$  and the two-sided  $p$ -values with `variance_ratio(r, q)`.
  3. Compute the Chow–Denning statistic  $\max_q |z^*(q)|$  and decide at 5% with the critical value 2.49; its  $p$ -value is  $1 - (2\Phi(\text{CD}) - 1)^4$ , with  $\Phi$  the distribution function of the standard Normal distribution.
- **Report:** a table with one row per market and  $q$  ( $\widehat{VR}$ ,  $z$ ,  $z^*$ ,  $p$ ), the CD decision, and one sentence on why  $|z| > |z^*|$
- Notebook: section B1

## B1 [Solved]: Solution (1/3) — S&P 500, $q = 2$ , Step by Step

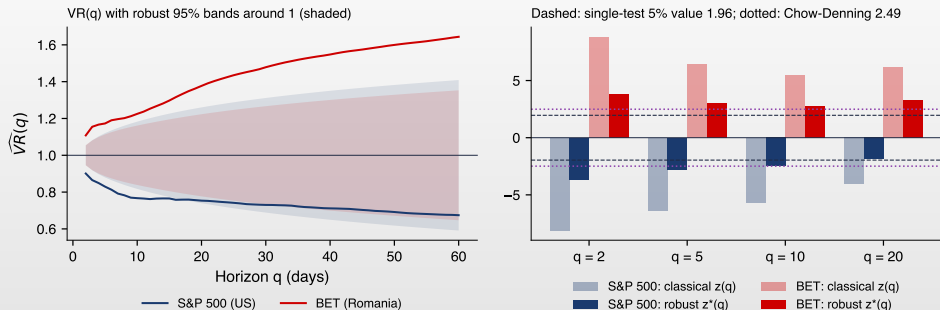
- ▣ Returns in %:  $T = 6\,714$ ,  $\hat{\mu} = 0.0247$ ; one-day variance  $\hat{\sigma}_a^2 = \frac{1}{T-1} \sum_t e_t^2 = 1.4714$
- ▣ 6 713 overlapping sums  $r_t + r_{t-1} - 2\hat{\mu}$ ; divisor  $m = q(T - q + 1)(1 - q/T) = 13\,422$ ;  $\hat{\sigma}_c^2(2) = 1.3256$
- ▣  $\widehat{VR}(2) = 1.3256/1.4714 = 0.9009$
- ▣ Classical:  $SD = \sqrt{2(2q-1)(q-1)/(3qT)} = \sqrt{1/T} = 0.0122$ , so  $z(2) = (-0.0991)/0.0122 = -8.12$
- ▣ Robust:  $\hat{\delta}_1 = T \sum_t e_t^2 e_{t-1}^2 / (\sum_t e_t^2)^2 = 4.966$  (it would be about 1 under i.i.d. returns); for  $q = 2$ ,  $\hat{\theta}(2) = \hat{\delta}_1$ 
  - ▶  $z^*(2) = \sqrt{T} (\widehat{VR}(2) - 1) / \sqrt{\hat{\theta}(2)} = 81.94 \times (-0.0991) / \sqrt{4.966} = -3.64$ ,  $p = 0.0003$
- ▣ The robust statistic is  $1/\sqrt{\hat{\delta}_1}$  times the classical one: volatility clustering makes the effective sample about 4.966 times smaller

## B1 [Solved]: Solution (2/3) — Both Markets, Four Horizons

|         | $z(2)$ | $z^*(2)$ | $z(5)$ | $z^*(5)$ | $z(10)$ | $z^*(10)$ | $z(20)$ | $z^*(20)$ |
|---------|--------|----------|--------|----------|---------|-----------|---------|-----------|
| S&P 500 | -8.12  | -3.64    | -6.37  | -2.77    | -5.65   | -2.50     | -4.06   | -1.86     |
| BET     | 8.83   | 3.84     | 6.44   | 2.98     | 5.44    | 2.73      | 6.19    | 3.31      |

- ▣  $\widehat{VR}(q)$  for  $q = 2, 5, 10, 20$ : S&P 500 0.901, 0.830, 0.767, 0.753; BET 1.108, 1.173, 1.225, 1.377
- ▣ Chow–Denning: S&P 500 3.64 ( $p = 0.0011$ ), BET 3.84 ( $p = 0.0005$ ): both reject RW3 jointly

## B1 [Solved]: Solution (3/3) — Chart and Interpretation



- Interpretation: reversal in the S&P 500, persistence in BET; the robust evidence is two to three times weaker, and a rejection is not a trading rule (costs, risk, stability: B7)

## B7 [Proposed]: Is the S&P 500 Reversal a Crisis Effect? — Task

- **Question:** does the S&P 500 reversal of B1 survive without the two crisis periods?
- **Data**
  - ▶ S&P 500, daily log returns 2000–2026,  $T = 6\,714$ ; crisis days, fixed in advance: 1 Sep 2008 – 30 Jun 2009 and 15 Feb – 30 Jun 2020
  - ▶ Removing a period leaves contiguous segments. A lag pair  $(r_t, r_{t-1})$  with  $t = 1$  Jul 2009 and  $t - 1 = 29$  Aug 2008 would join two days nine months apart: it is forbidden; so is any 5-day sum that spans a gap
- **Tasks**
  1. On all days compute  $\hat{\rho}_1$ , the i.i.d.  $t = \hat{\rho}_1 \sqrt{T}$ , the robust  $t = \hat{\rho}_1 / \sqrt{\hat{\tau}_1}$  and  $z^*(5)$ .
  2. Remove the crisis days and repeat within the retained segments; report the kept returns, the valid lag pairs and the valid 5-day sums.
  3. Give robust 95% intervals for  $\rho_1$  in both samples and decide at 5%.
- **Report**
  - ▶ a two-row table and one sentence
  - ▶ by how much does the estimate fall?
  - ▶ is removing crisis days legitimate?
- **Model:** B1 [Solved]
- **Notebook:** section B7

## C2 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent by a student to an AI assistant: “Test whether the S&P 500 (daily closes, 2000–2026) follows a random walk with the variance ratio. Give Python code and interpret the result”

```

1  r = np.log(sp).diff().dropna()          # sp: S&P 500 closes, as in B1
2  T = len(r)
3  def vr(q):
4      rq = r.rolling(q).sum().dropna()    # overlapping q-day returns
5      return rq.var() / (q * r.var())
6  q = 5
7  z = (vr(q) - 1) / np.sqrt(2 * (2*q - 1) * (q - 1) / (3 * q * T))
8  p = 2 * stats.norm.sf(abs(z))

```

- **The AI's interpretation:** (a) “ $VR(2) = 0.901$ ,  $VR(5) = 0.830$ ,  $z(5) = -6.37$ ,  $p < 0.001$ : the random walk is rejected”; (b) “since  $VR(2) = 1 + 2\rho_1$ ,  $\rho_1 \approx -0.050$ ”; (c) “ $VR < 1$  means momentum: buy the S&P 500 after up days”

## C2 [Proposed]: Critique an AI Answer — Task

- ▣ **Question:** which claims of the AI answer does its code actually support?
- ▣ **Data:** the code and interpretation on the previous slide; the same S&P 500 returns as B1
- ▣ **Tasks**
  1. Audit four points in a table (line number, claim, correct or wrong, how you check): the estimator convention, the null and its standard error, the VR identity, the economic interpretation.
  2. Say which reported numbers the code computes (it computes only  $q = 5$ ).
  3. Give two numerical checks:  $\hat{\rho}_1$  computed directly, and  $z^*(5)$  from `variance_ratio`.
  4. Write the corrected interpretation in two sentences.
- ▣ **Report:** the audit table, the two checks, the corrected interpretation; log each error as in `AI_ERRORS.md`
- ▣ **Model:** A1, B1 [Solved]
- ▣ **Notebook:** section B1

## B5 [Solved]: The Turn-of-the-Month Effect with HAC Errors — Task

- ▣ **Question:** do returns on turn-of-the-month days exceed those on other days?
- ▣ **Data**
  - ▶ S&P 500 and BET, daily log returns 2000–2026 in basis points,  $r_t = 10\,000 \ln(P_t/P_{t-1})$ , each on its own trading calendar; the incomplete last month (September 2026) is dropped
  - ▶  $\text{TOM}_t = 1$  on the last trading day of a month and the first three trading days of the next [Ariel \(1987\)](#), 0 otherwise; example: 31 Jul and 3, 4, 5 Aug 2026
- ▣ **Tasks**
  1. Build  $\text{TOM}_t$  from the trading calendar (rank of the day within its month, from the start and from the end).
  2. Estimate  $r_t = a + b \text{TOM}_t + u_t$  by OLS;  $\hat{a}$  is the mean on other days,  $\hat{a} + \hat{b}$  the mean on TOM days.
  3. Compute the OLS and the Newey–West HAC standard errors of  $\hat{b}$  with 5 lags [Newey & West \(1987\)](#), both  $t$ -statistics and the HAC  $p$ -value.
  4. Convert  $\hat{b}$  into an approximate extra log return per month (4 TOM days), before costs.
- ▣ **Report:** a table (share of TOM days,  $\hat{a}$ ,  $\hat{b}$ , both SEs and  $t$ ), a 95% interval chart, one sentence on why OLS and HAC differ
- ▣ **Notebook:** section B5

## B5 [Solved]: Solution (1/3) — Design Matrix

| date       | $r_t$ (bp) | 1 | TOM <sub>t</sub> |
|------------|------------|---|------------------|
| 2026-07-27 | +1.6       | 1 | 0                |
| 2026-07-28 | +21.0      | 1 | 0                |
| 2026-07-29 | -152.8     | 1 | 0                |
| 2026-07-30 | +164.7     | 1 | 0                |
| 2026-07-31 | +69.8      | 1 | 1                |
| 2026-08-03 | +146.8     | 1 | 1                |
| 2026-08-04 | +177.4     | 1 | 1                |
| 2026-08-05 | -16.8      | 1 | 1                |
| 2026-08-06 | -17.6      | 1 | 0                |
| 2026-08-07 | +61.7      | 1 | 0                |

- S&P 500 rows around the July/August 2026 boundary: the regressors are a column of ones and the TOM dummy
- OLS:  $\hat{b} = \bar{r}_{\text{TOM}} - \bar{r}_{\text{other}}$ ; S&P 500  $5.1 - 1.9 = 3.2$  bp; BET  $20.4 - 3.1 = 17.3$  bp
- HAC:  $\widehat{\text{Var}}(\hat{\beta}) = (X'X)^{-1} \hat{S} (X'X)^{-1}$ ,  $\hat{S} = \sum_{|j| \leq 5} (1 - \frac{|j|}{6}) \sum_t x_t \hat{u}_t \hat{u}_{t-j} x'_{t-j}$ : it lets the residual variance change and neighbouring days correlate
- In code: `sm.OLS(r, X).fit(cov_type='HAC', cov_kwds={'maxlags': 5})`

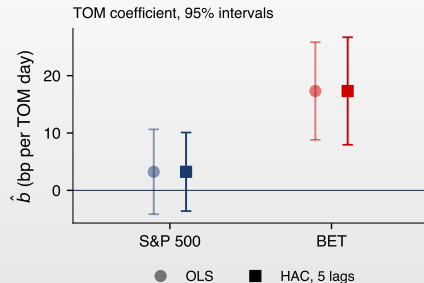
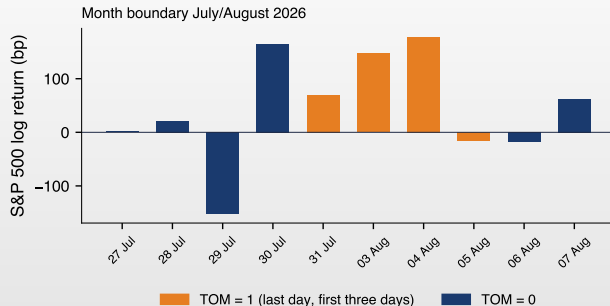
 MFM\_ch2\_calendar\_anomalies

## B5 [Solved]: Solution (2/3) — Estimates

|         | TOM share | $\hat{a}$ (bp) | $\hat{b}$ (bp) | $SE_{OLS}$ | $t_{OLS}$ | $SE_{HAC}$ | $t_{HAC}$ (p) |
|---------|-----------|----------------|----------------|------------|-----------|------------|---------------|
| S&P 500 | 19.1%     | 1.9            | 3.2            | 3.8        | 0.86      | 3.5        | 0.93 (0.35)   |
| BET     | 19.2%     | 3.1            | 17.3           | 4.3        | 3.99      | 4.8        | 3.62 (0.0003) |

- BET: HAC 95% interval [8.0, 26.7] bp; the HAC SE exceeds the OLS SE (clustering and  $\rho_1 > 0$ ),  $t$  falls from 3.99 to 3.62, still a rejection
- S&P 500:  $[-3.6, 10.1]$  bp contains 0
- $4\hat{b} \approx 69$  bp (BET): an approximate extra monthly log return relative to the other-day mean, before costs, not a strategy return

## B5 [Solved]: Solution (3/3) — Chart and Interpretation



- Interpretation: one effect in one market was picked among many: correct for the effects tried (B6) before looking for a mechanism (salary and pension inflows)

## B6 [Proposed]: Fifteen Calendar Tests and Holm — Task (1/2)

- **Question:** which of fifteen calendar effects survive a correction for multiple testing?
- **Data:** the 15  $p$ -values in the table below: 5 markets  $\times$  3 effects (day of week: Wald test of equal means, HAC 5 lags; January: monthly returns, HAC 3 lags; TOM: as in B5), 2000–2026

| Market        | Effect                    | $p$    | Market         | Effect                    | $p$    |
|---------------|---------------------------|--------|----------------|---------------------------|--------|
| S&P 500 (US)  | day of week (equal means) | 0.826  | Euro Stoxx 50  | turn of the month         | 0.227  |
| S&P 500 (US)  | January                   | 0.426  | WIG20 (Poland) | day of week (equal means) | 0.226  |
| S&P 500 (US)  | turn of the month         | 0.354  | WIG20 (Poland) | January                   | 0.686  |
| BET (Romania) | day of week (equal means) | 0.305  | WIG20 (Poland) | turn of the month         | 0.084  |
| BET (Romania) | January                   | 0.137  | BUX (Hungary)  | day of week (equal means) | 0.933  |
| BET (Romania) | turn of the month         | 0.0003 | BUX (Hungary)  | January                   | 0.024  |
| Euro Stoxx 50 | day of week (equal means) | 0.632  | BUX (Hungary)  | turn of the month         | 0.0003 |
| Euro Stoxx 50 | January                   | 0.774  |                |                           |        |

## B6 [Proposed]: Fifteen Calendar Tests and Holm — Task (2/2)

### □ Tasks

1. Count the raw rejections at 5% and compute the FWER of 15 independent tests.
2. Apply Bonferroni: reject if  $p_i < 0.05/15$ .
3. Apply Holm: sort the  $p$ -values, compare  $p_{(j)}$  with  $0.05/(16 - j)$  and stop at the first failure.

□ **Report:** the sorted table with ranks, thresholds and three decision columns (raw, Bonferroni, Holm), and one sentence on BUX January

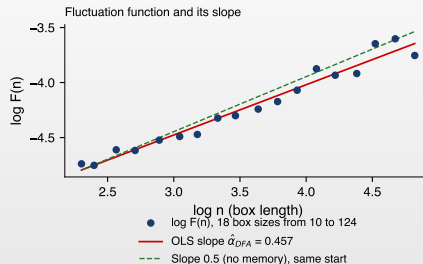
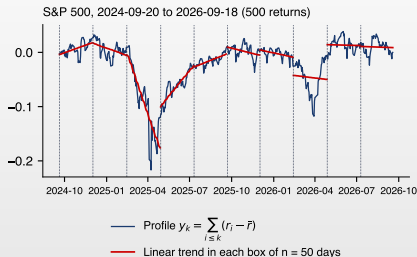
□ **Model:** B5 [Solved]

□ **Notebook:** section B6

## B3 [Solved]: Rolling DFA on the S&P 500 — Task

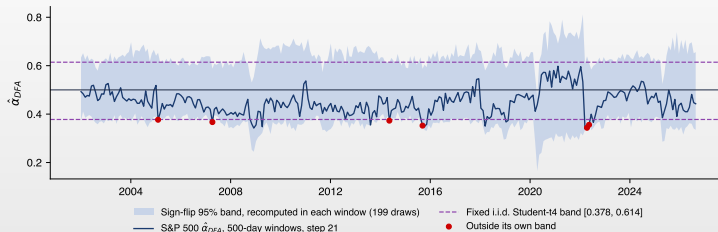
- **Question:** does the S&P 500 DFA exponent ever leave a null band that allows volatility clustering?
- **Data**
  - ▶ S&P 500, daily log returns 2000–2026; windows of 500 returns moved by 21 days (296 windows); function `dfa_hurst(x)`
  - ▶ DFA grid: 18 box lengths spaced evenly in  $\log n$  from 10 to about 125 (a quarter of the window; 124 after rounding), rounded to integers, duplicates removed; the remainder  $T - kn$  of the profile is discarded; random seed = 42 + index of the last day of the window
- **Tasks**
  1. For the last window (2024-09-20 to 2026-09-18) build the profile, detrend it in boxes and fit  $\log F(n)$  on  $\log n$ .
  2. Compute  $\alpha_{DFA}$  in every window.
  3. In each window build a sign-flip band: 199 series  $e_t \eta_t$ ,  $\eta_t = \pm 1$  with probability 1/2; keep the 2.5% and 97.5% quantiles of  $\alpha_{DFA}$ .
  4. Compute the share of windows outside their own band and outside one fixed i.i.d. Student- $t_4$  band (300 simulations of length 500).
- **Report:** the one-window fit, the rolling chart, both shares, and one sentence on what the band assumes
- **Notebook:** section B3

## B3 [Solved]: Solution (1/2) — One Window, Step by Step



- Profile  $y_k = \sum_{i \leq k} (r_i - \bar{r})$  (left); in each box of length  $n$  fit a line and keep the residuals;  $F(n)$  = root mean square of the residuals
- 18 box lengths from 10 to 124; the OLS slope of  $\log F(n)$  on  $\log n$  is  $\hat{\alpha}_{DFA} = 0.457$  (right), close to 0.5

## B3 [Solved]: Solution (2/2) — All Windows



- Outside their own band: 2.0% of 296 windows; outside the fixed Student- $t_4$  band [0.378, 0.614]: 6.4%
- Mean  $\hat{\alpha}_{DFA} = 0.450$ ; minimum 0.341 (window ending 2022-03-15), maximum 0.598 (2021-02-12); the band is widest (0.500) in the window ending 2022-02-11
- Interpretation: the band is a pointwise benchmark (signs symmetric and independent given  $|e_t|$ ), not a confidence interval; overlapping windows are correlated, so the shares describe, they do not test stability; no memory beyond what chance would produce

## B4 [Proposed]: Adaptive Efficiency of Bitcoin — Task

- **Question:** has Bitcoin moved towards efficiency since its early years?
- **Data:** Bitcoin (USD) closing prices, daily log returns 7 days a week from 1 Jan 2011 to 18 Sep 2026 (earlier than the lecture's 2014 start, to include the first bubbles); windows of 365 days moved by 30 days;  $H_0$ : RW3 [Lo \(2004\)](#)
- **Tasks**
  1. Compute the robust  $z^*(5)$  in each window.
  2. In each window build a sign-flip band for  $z^*(5)$  with 999 replications (random seed = 42 + index of the last day of the window) and keep its 2.5% and 97.5% quantiles.
  3. Count the windows outside their band and outside  $\pm 1.96$  and compare with  $0.05 \times$  the number of windows.
  4. Compare with [Urquhart \(2016\)](#): 2010–2016, a battery of classical tests (Ljung–Box, runs, Bartels, automatic VR, BDS, R/S).
- **Report:** the chart, the flagged windows with  $z^*$  and band, both counts, and one sentence: what does “no detected trend” mean?
- **Model:** B3, B1 [Solved]
- Notebook: section B4

## After the Lecture: What and When

- These two tasks use tools that [Chapter 2](#) introduces: work on them after the lecture, before the next seminar
  - ▶ B3 [Solved]: detrended fluctuation analysis (DFA) on rolling windows, with a sign-flip (wild bootstrap) null band
  - ▶ B4 [Proposed]: the robust  $z^*(5)$  of B1 on rolling windows of Bitcoin, with the same kind of band
- The solution of B4 is discussed at the next seminar

## Extension Packs: What Each One Needs

| Pack              | Exercises                                                       | Prerequisites                  |
|-------------------|-----------------------------------------------------------------|--------------------------------|
| 1. Theory         | A2 (null distribution), A3 [Solved] (runs), A4 (Stambaugh bias) | A1, B1                         |
| 2. Many markets   | B2                                                              | B1                             |
| 3. Predictability | B8                                                              | A4, B5                         |
| 4. Event study    | B9                                                              | Seminar 0, B7; B5              |
| 5. Project seeds  | C1, C3                                                          | B1, B3; the lecture case study |

- Each exercise has its own notebook section; do the packs in order, the solutions of the proposed ones are discussed in class

## A2 [Proposed]: The Variance Ratio and Its Null Distribution — Task

### ▣ Questions

- ▶ why is  $\sqrt{T}(\widehat{\text{VR}}(q) - 1)$  approximately Normal with variance  $2(2q - 1)(q - 1)/(3q)$ ?
- ▶ when does this fail?

- ▣ **Data:**  $r_t$  covariance-stationary, variance  $\gamma_0$ , autocovariances  $\gamma_k = \text{Cov}(r_t, r_{t-k})$ ,  $\rho_k = \gamma_k/\gamma_0$ ; under RW1  $\sqrt{T}(\hat{\rho}_1, \dots, \hat{\rho}_{q-1})$  tends to  $N(0, I_{q-1})$ ,  $I_{q-1}$  the identity matrix (independent, unit variances); S&P 500:  $T = 6\,714$ ,  $\widehat{\text{VR}}(2) = 0.901$

### ▣ Tasks

1. For  $q = 3$ , write  $\text{Var}(r_t + r_{t-1} + r_{t-2})$  as the sum of the nine covariances and count how many times  $\gamma_0$ ,  $\gamma_1$ ,  $\gamma_2$  appear.
2. Generalise: show that  $\text{VR}(q) = 1 + \sum_{k=1}^{q-1} a_k \rho_k$  with  $a_k = 2(1 - k/q)$ .
3. Under RW1, with  $a = (a_1, \dots, a_{q-1})'$ , compute  $a'la = \sum_k a_k^2$  and simplify it to  $2(2q - 1)(q - 1)/(3q)$ ; evaluate it for  $q = 2, 5, 20$ .
4. Compute  $z(2)$  for the S&P 500 and explain why  $I$  must be replaced by  $\text{diag}(\delta_k)$ , the diagonal matrix of the robust variances  $\delta_k$ , under RW3.

- ▣ **Report:** the counting, the variance in two columns (asymptotic, of  $\sqrt{T}(\widehat{\text{VR}} - 1)$ ; finite-sample, of  $\widehat{\text{VR}}$ : divide by  $T$ ) and  $z(2)$

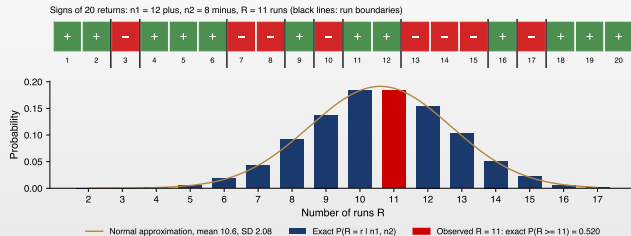
- ▣ **Model:** A1 [Solved]

- ▣ **Notebook:** section A2

## A3 [Solved]: The Runs Test, Step by Step — Task

- ▣ **Question:** are 20 signs consistent with independence?
- ▣ **Data:** signs of 20 consecutive daily returns, no zeros: ++-+++--+-+---+---+; under  $H_0$  the signs are independent with a constant probability of a positive day [Wald & Wolfowitz \(1940\)](#)
- ▣ **Tasks**
  1. Number the signs 1 to 20, mark every sign change and count  $n_1$ ,  $n_2$  and the runs  $R$ .
  2. Compute  $\mathbb{E}[R] = 2n_1n_2/n + 1$  and  $\text{Var}[R] = 2n_1n_2(2n_1n_2 - n)/(n^2(n - 1))$ .
  3. Compute the approximate  $z$  and its two-sided  $p$ -value.
  4. Compare with the exact conditional distribution of  $R$  given  $n_1$  and  $n_2$ .
- ▣ **Report:**  $n_1$ ,  $n_2$ ,  $R$ ,  $\mathbb{E}[R]$ ,  $\text{Var}[R]$ ,  $z$ , both  $p$ -values and what too few or too many runs would mean
- ▣ **Notebook:** section A3

## A3 [Solved]: Solution



- (1) Runs: ++ | - | +++ | -- | + | - | ++ | --- | + | - | +++, so  $n_1 = 12$ ,  $n_2 = 8$ ,  $R = 11$  (10 sign changes)
- (2)  $\mathbb{E}[R] = 2 \cdot 12 \cdot 8/20 + 1 = 10.60$ ;  $\text{Var}[R] = 192 \cdot 172/(400 \cdot 19) = 4.345$ ; (3) approximate  $z = 0.40/2.085 = 0.19$ ,  $p = 0.85$
- (4) Exact:  $P(R = 11) = 0.183$ ,  $P(R \geq 11) = 0.520$ ,  $P(R \leq 11) = 0.663$ , so the exact two-sided  $p$  is 1. Interpretation: do not reject independence; too few runs would mean persistent signs, too many alternating signs

## A4 [Proposed]: The Stambaugh Bias — Task

▣ **Question:** why does a persistent valuation ratio look like a significant predictor too often?

▣ **Data**

- ▶  $x_t$ : the log dividend–price ratio, very persistent; when prices rise,  $x$  falls, so the innovations of returns and of  $x$  are negatively correlated
- ▶ System:  $r_{t+1} = \alpha + \beta x_t + u_{t+1}$ ,  $x_{t+1} = \theta + \rho x_t + v_{t+1}$ ; write  $u = \phi v + e$ ,  $\phi = \sigma_{uv} / \sigma_v^2$ , with  $\sigma_{uv} = \text{Cov}(u, v)$ ,  $\sigma_u$ ,  $\sigma_v$  the shock SDs and  $e$  the part of  $u$  unrelated to  $v$ ; Kendall:  $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$  [Stambaugh \(1999\)](#)
- ▶ Illustrative annual inputs (as fractions, not in %):  $\rho = 0.98$ ,  $T = 60$  years,  $\text{corr}(u, v) = -0.9$ ,  $\sigma_u = 0.17$ ,  $\sigma_v = 0.14$

▣ **Tasks**

1. With demeaned  $\tilde{x}_t = x_t - \bar{x}$ , show  $\hat{\beta} - \beta = \phi(\hat{\rho} - \rho) + \sum \tilde{x}_t e_{t+1} / \sum \tilde{x}_t^2$  and derive  $\mathbb{E}[\hat{\beta} - \beta]$ .
2. Compute  $\phi$ ,  $\mathbb{E}[\hat{\rho} - \rho]$  and the bias of  $\hat{\beta}$ .
3. Compute  $\text{SE}(\hat{\beta}) \approx \sigma_u / (\sigma_x \sqrt{T})$ ,  $\sigma_x = \sigma_v / \sqrt{1 - \rho^2}$ , and the bias in standard errors.

▣ **Report:** the derivation,  $\phi$ , the bias, the SE, their ratio, and what a positive bias does to the false-rejection rate

▣ **Model:** A1 [Solved]

▣ **Notebook:** section A4

## B2 [Proposed]: BET-TR and Emerging Markets — Task

- ▣ **Question:** are emerging markets less efficient as a group?
- ▣ **Data:** BET-TR (from 2014) and WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai, closing prices, 2000–2026, each on its own calendar
- ▣ **Tasks**
  1. Start with BET and BET-TR on the same dates (BET-TR's period): classical  $z(2)$ , robust  $z^*(2)$ , Chow–Denning.
  2. Repeat for the eight other indices and report three decisions separately: classical  $q = 2$ , robust  $q = 2$ , robust joint over four horizons.
  3. Before any group-wide claim, correct the joint  $p$ -values for the nine markets tested (Holm).
- ▣ **Report:** one table (one row per market), the matched BET comparison and one sentence: dividends or sample?
- ▣ **Model:** B1 [Solved]
- ▣ **Notebook:** section B2

## B8 [Proposed]: Predictive Regression and an Out-of-Sample Test — Task (1/2)

- **Question:** does the dividend–price ratio predict S&P 500 excess returns, in and out of sample?
- **Data**
  - ▶ Monthly table 1934–2026 ( $T = 1\,111$ ):  $P$  = Shiller S&P Composite monthly average price (to 2026-06), extended with S&P 500 month-end closes;  $D$  = trailing 12-month dividends (Shiller, interpolated from quarterly totals), extended with the dividends of the SPY fund (ETF on the S&P 500), inferred from the gap between its closing price and its dividend-adjusted price, spliced on the common period
  - ▶  $r_{t+1} = \ln((P_{t+1} + D_{t+1}/12)/P_t) - \ln(1 + \text{TB3MS}_t/1\,200)$  (FRED, known at  $t$ ); predictor  $dp_t = \ln D_t - \ln P_t$ , used one month later

## B8 [Proposed]: Predictive Regression and an Out-of-Sample Test — Task (2/2)

### ▣ Tasks

1. Estimate  $r_{t+1} = \alpha + \beta dp_t + u_{t+1}$  by OLS with Newey–West errors (12 lags); estimate the AR(1) of  $dp$  and  $\text{corr}(\hat{u}, \hat{v})$ ; correct  $\hat{\beta}$  for the Stambaugh bias (A4).
2. From January 1970, at each month re-estimate on data up to the previous month and forecast; the benchmark is the historical mean of the same data; compute  $R_{OS}^2$  [Welch & Goyal \(2008\)](#) (pseudo out-of-sample: dividends are interpolated).
3. Compute the Clark–West  $t$  (HAC, 12 lags, one-sided) [Clark & West \(2007\)](#); impose the Campbell–Thompson restrictions (slope sign, non-negative forecast) [Campbell & Thompson \(2008\)](#).

▣ **Report:**  $\hat{\beta}$ ,  $t$ ,  $\hat{\rho}$ , the bias,  $R_{OS}^2$ , the Clark–West  $t$ , and why Diebold–Mariano is the wrong test for nested models

▣ **Model:** A4, B5

▣ **Notebook:** section B8

## B9 [Proposed]: The Bank Tax, TLV and BRD Jointly — Task (1/2)

- ▣ **Question:** how much of the bank-tax fall survives proper event-study inference?
- ▣ **Data**
  - ▶ Banca Transilvania (TLV) and BRD (dividend- and split-adjusted prices) and BET (closing prices), prices aligned on common trading days, then log returns
  - ▶ Timeline: tax announced on the evening of 18 Dec 2018; day 0 = 19 Dec; estimation window days  $[-250, -11]$  ( $T_0 = 240$ ); event window  $[0, +5]$
  - ▶  $X_0$  ( $240 \times 2$ ) and  $X_e$  ( $6 \times 2$ ): the  $[1, r_{m,t}]$  rows of the estimation and event windows;  $\iota$ : a  $6 \times 1$  vector of ones;  $I_6$ : the  $6 \times 6$  identity matrix;  $N$ : number of firms;  $\bar{r}$ : average correlation of their residuals

## B9 [Proposed]: The Bank Tax, TLV and BRD Jointly — Task (2/2)

### ▣ Tasks

1. For TLV fit the market model on  $[-250, -11]$ , compute  $AR_0$ , the naive  $t = AR_0 / \hat{\sigma}_\varepsilon$  and the  $t$  with the prediction-error variance; repeat for BRD.
2. For the equal-weighted portfolio of the two banks fit the market model to its own returns and compute its  $AR_0$  and  $t$ ; compare with the  $t$  that assumes independent banks.
3. Compute  $CAR[0, +5]$  and  $CAR[+1, +5]$  with  $\text{Var}(CAR) = \iota' \hat{\sigma}_\varepsilon^2 [\mathbf{I}_6 + X_e(X_0' X_0)^{-1} X_e'] \iota$  (adapted to the window used).
4. Compute the residual correlation  $\bar{r}$  and the Kolari–Pynnönen factor  $\sqrt{(1 - \bar{r}) / (1 + (N - 1)\bar{r})}$  [Kolari & Pynnönen \(2010\)](#), a correction for cross-sectional tests with many firms.

▣ **Report:** one table (TLV, BRD, portfolio), the AR/CAR chart, and whether the reversal contradicts semi-strong efficiency; note that BET contains both banks

▣ **Model:** Seminar 0, B7; B5

▣ **Notebook:** section B9

## C1 [Proposed]: BVB Efficiency after the 2020 FTSE Upgrade — Task (1/2)

- **Question:** did BET's lag-1 autocorrelation move towards zero after the upgrade, relative to WIG20 and BUX?
- **Data**
  - ▶ **FTSE Russell (2019):** announced 26 Sep 2019, effective 21 Sep 2020 (Frontier to Secondary Emerging); background: [Pele & Mazurencu-Marinescu \(2012\)](#)
  - ▶ BET, WIG20, BUX, closing prices; before: 21 Sep 2015 – 20 Sep 2020; after: 21 Sep 2020 – 21 Sep 2025; data after 21 Sep 2025 only in a separately reported extension

## C1 [Proposed]: BVB Efficiency after the 2020 FTSE Upgrade — Task (2/2)

### ▣ Tasks

1. For each market compute  $\hat{\rho}_1$  with robust SE and  $z^*(5)$  in both windows.
2. Build a 95% interval for the change in  $\hat{\rho}_1$  by a moving-block bootstrap, independent within each period (blocks of 20 days, 999 replications, seed 42) [Künsch \(1989\)](#).
3. Plot rolling 250-day  $\hat{\rho}_1$  with robust intervals for the three markets, marking the announcement, the effective date and the end of the formal window.
4. List the confounders (COVID-19, the 2023 Hidroelectrica listing, liquidity) and say what a control-adjusted contrast would need.

▣ **Report:** a Quantlet folder and one slide: conclusion, main limitation, link with the AMH [Lo \(2004\)](#); judge movement towards zero, not the sign

▣ **Model:** B1, B3 [Solved]

▣ **Notebook:** section C1

## C3 [Proposed]: Big Data and Efficiency, In and Out of Sample — Task (1/2)

- **Question:** with 238 predictors, is in-sample predictability of industry returns matched out of sample?
- **Data**
  - ▶ Lecture 2 case study [Martin & Nagel \(2022\)](#): rational learning makes in-sample tests reject although out-of-sample returns are unpredictable; replicate the design of Sec. 7 of their paper
  - ▶ 49 industry portfolios, [Kenneth French Data Library](#), value-weighted monthly returns, July 1926 to the last month; missing values −99.99 and −999 dropped month by month; returns divided by 100
  - ▶ Dimensions: in month  $t$ ,  $X_t$  is  $N_t \times 238$  (returns and squared returns at lags 2–120),  $N_t$  the industries with a complete 120-month history;  $h_t$  the  $238 \times 1$  coefficients

## C3 [Proposed]: Big Data and Efficiency, In and Out of Sample — Task (2/2)

### ▣ Tasks

1. Demean the returns and each predictor across industries every month, scale each predictor to unit cross-sectional SD, give each month equal weight.
2. Estimate pooled OLS  $h_t$  over the 240 months ending in  $t$  (data up to  $t$  only); compute  $roos_{t+1} = \frac{1}{N} h_t' X_{t+1}' r_{t+1}$  and  $rs$  over the same window; state the first forecast month.
3. Average  $roos$  over 10-year windows with descriptive  $\pm 2$  SE bands (as in Fig. 8 of the paper).
4. Fit a ridge from January 1971 with standardised predictors; choose the penalty on a log grid by leave-one-year-out CV (it may use later years: a validation, not a forecast); plot the coefficients by lag.

▣ **Report:** the chart of step 3, early and late means of  $rs$  and  $roos$  with SE, the coefficient plot, and why 49 portfolios can differ from thousands of stocks

▣ **Model:** B5, B8

▣ **Notebook:** section C3 (lecture Quantlet MFM\_ch2\_big\_data)

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